

Lab 2: Agile Backlog Creation & Sprint Simulation

Reflection Report

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Date	8 September 2026
Jira Project	JiraLab2 (JIR)

1. Project Overview

This lab converts the functional requirements identified in Lab 1 into a structured Agile backlog using Jira, and simulates a single sprint to demonstrate the Scrum workflow end-to-end. The scenario used is a courier/delivery dispatch application. The Lab 1 requirements were organised into four Epics:

- Courier Dispatch Management — matching, assigning, and managing rider availability for delivery requests.
- Delivery Tracking & Verification — real-time tracking, status notifications, and OTP-based delivery verification.
- Delivery Route Optimization — route optimization, alternative routing, and ETA estimation.
- Payment Processing & Authorization — authorizing payment before a rider is dispatched.

Each Epic was broken down into User Stories written in the standard “As a / I want / So that” format, prioritized, and estimated using Fibonacci story points. All ten stories were planned and executed within a single sprint (JIR Sprint 1, 4–8 September 2026), per project instructions to complete the exercise in one sprint rather than the two suggested in the lab handout.

2. Backlog: Epics and User Stories

The screenshot below shows the full Jira backlog with all four Epics and their nested User Stories.

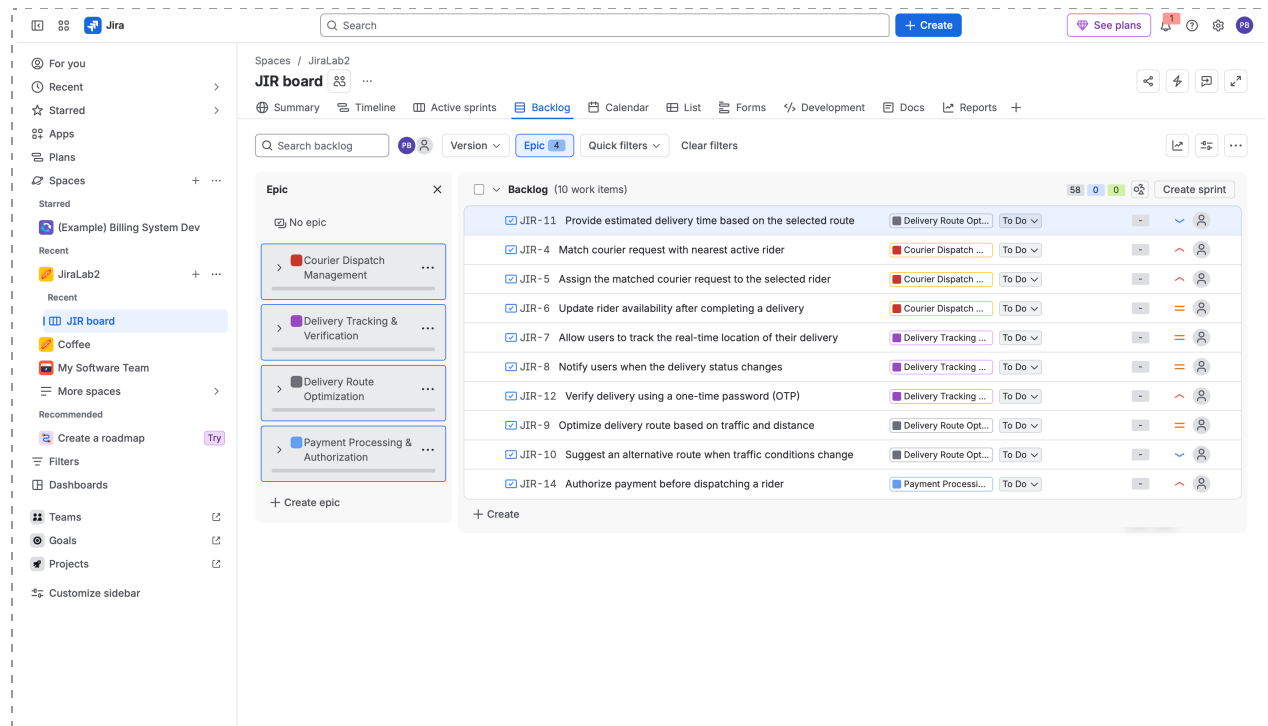


Figure 1: Jira Backlog showing Epics and User Stories

3. Story Point Assignments

Story points were assigned using the Fibonacci scale (1, 2, 3, 5, 8, 13) based on relative complexity, effort, and uncertainty for each story.

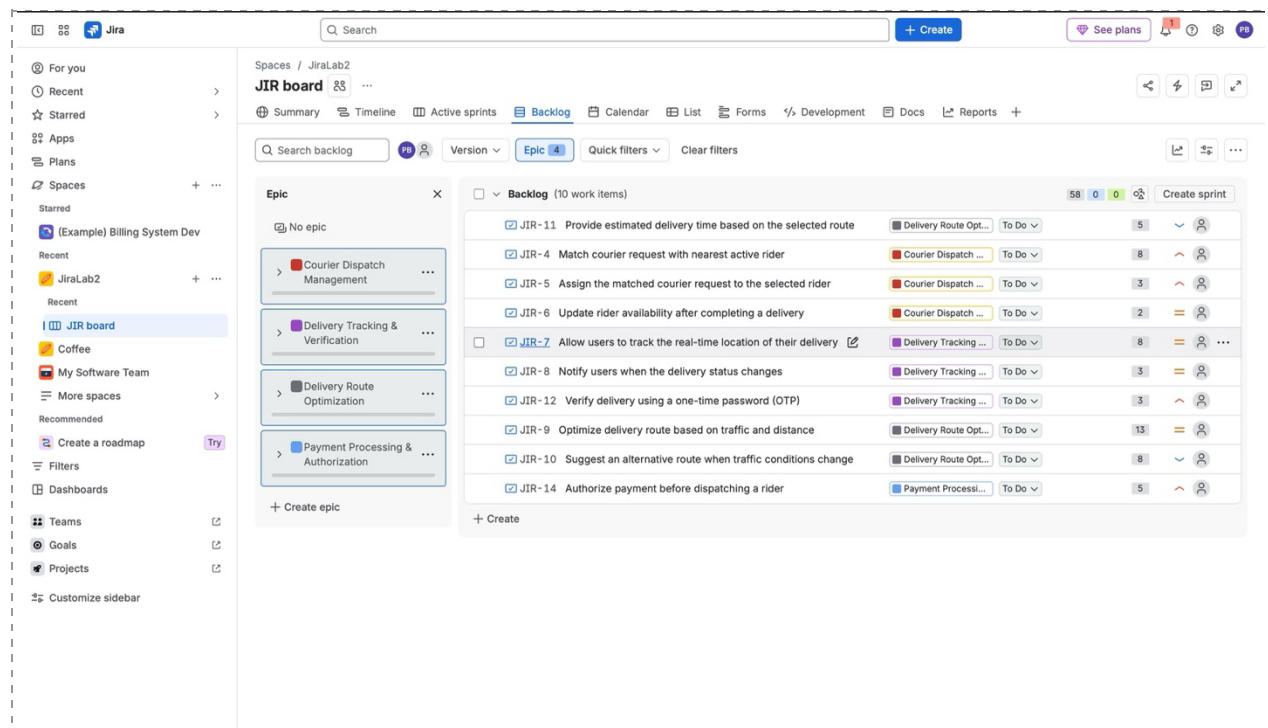


Figure 2: Story Point Assignments

4. Sprint Execution

Sprint 1 was run from 4 September to 8 September 2026, containing all 10 backlog stories. Stories were progressively moved from To Do → In Progress → Done to simulate real sprint execution across several intervals.

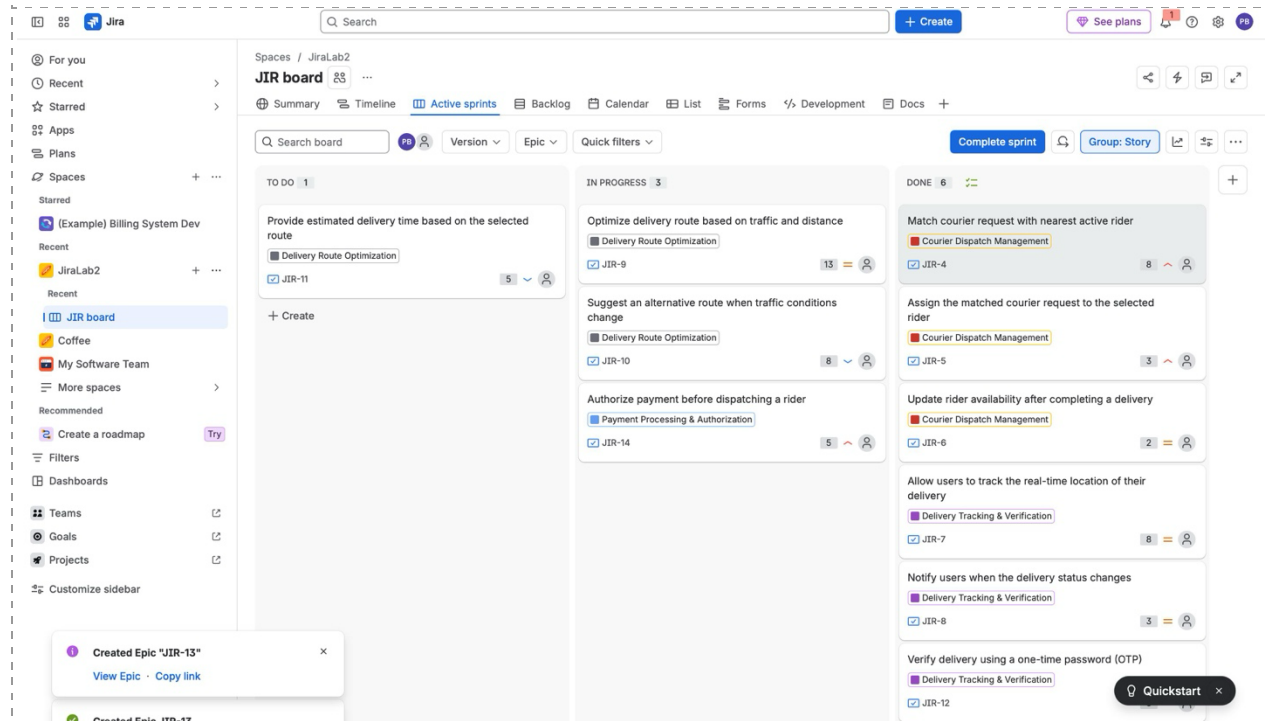


Figure 3: Sprint Board (Active Sprint view)

5. Burndown Chart

After completing the sprint, the Burndown Chart was generated from Reports → Burndown Chart, tracking remaining story points against the ideal guideline.

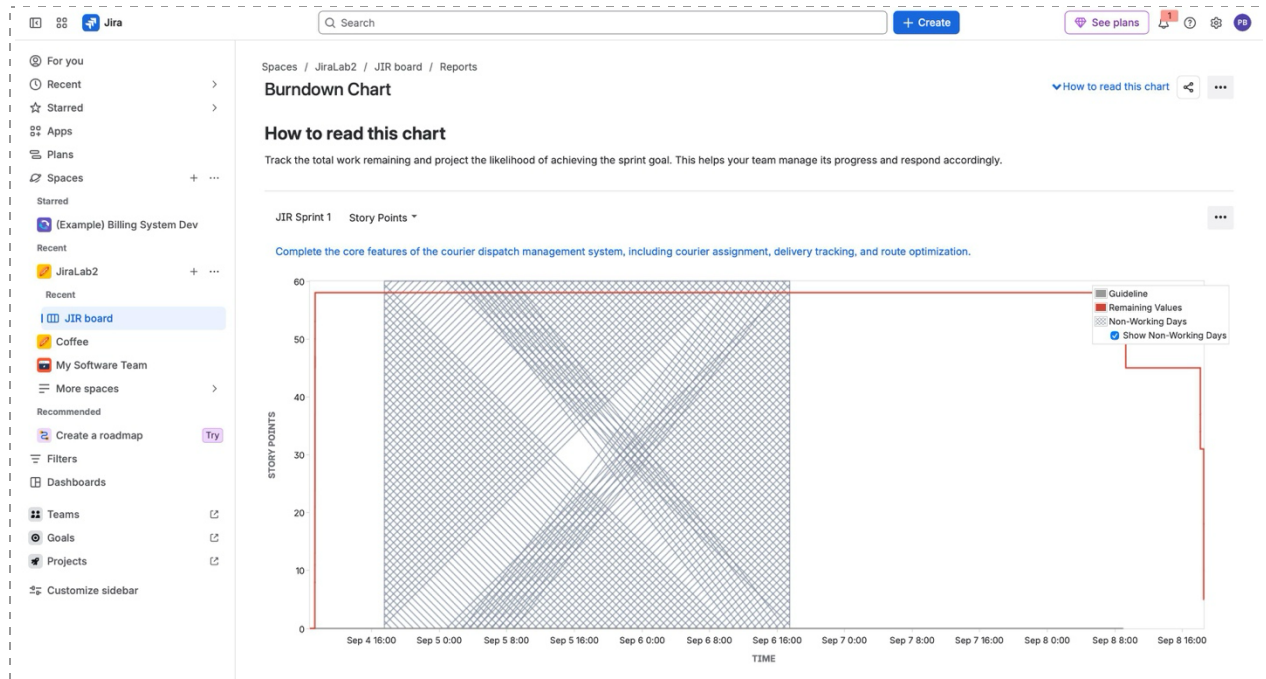


Figure 4: Sprint Burndown Chart

6. Reflection Questions

6.1 Did your estimations reflect the actual effort?

For the most part, yes. Higher-point stories like JIR-9 (Optimize delivery rAuthorized on traffic and distance, 13 pts) genuinely involve more complex logic — combining live traffic data, distance calculation, and route-selection algorithms — compared to a simple status-flag update like JIR-6 (Update rider availability, 2 pts), which is a single field change. The mid-range stories (JIR-4, JIR-7, JIR-10 at 8 pts) also felt appropriately estimated, since they involve real-time or conditional logic (proximity matching, live location tracking, dynamic rerouting) without the full complexity of the route optimization engine. One estimate that could be revisited is JIR-14 (Authorize payment, 5 pts) — in practice, payment gateway integrations often carry more edge-case risk (failed transactions, retries, timeouts) than a 5-point estimate suggests, so this might deserve an 8 in a real implementation.

6.2 Was your backlog well-prioritized?

Yes, the prioritization followed a clear logic: High priority was assigned to the core-gating stories — rider matching (JIR-4), rider assignment (JIR-5), OTP verification (JIR-12), and payment authorization (JIR-14) — since the delivery workflow cannot function correctly without these steps in place. This ordering held up well in practice: all four High-priority stories were among the first completed in the sprint, confirming the prioritization the experience but don't block the core flow. Low priority was given to enhancements — alternative route suggestions and ETA display — which are valuable but non-essential. This ordering held up well in practice: all four High-priority stories were among the first completed in the sprint, confirming the prioritization matched actual execution.

6.3 How did your simulated sprint align with your plan?

The sprint (JIR Sprint 1, 4–8 September 2026) was planned to complete all 10 backlog stories within a single sprint, per the assignment's requirement to use one sprint instead of the two suggested in the lab handout. By the end of the sprint window, 9 of the 10 stories were moved to Done, with JIR-11 (Provide estimated delivery time based on the selected route) left incomplete — it was still in the To Do column when the sprint was closed. This is a realistic outcome: lower-priority, enhancement-type stories are often the first to slip when a sprint is compressed, which is exactly what happened here since JIR-11 was one of the Low-priority route-optimization stories.

6.4 What insights did the burndown chart give about your team's capacity?

The burndown chart shows the remaining-values line staying flat near 58 points for most of the sprint before dropping sharply near the end, rather than decreasing steadily against the guideline. This indicates that work was completed in large batches rather than continuously throughout the sprint — a common pattern when a team (or in this case, an individual working the simulation) processes tasks in concentrated sessions instead of spreading effort evenly across the sprint calendar. Compared to the guideline line, the team was significantly behind for most of the sprint and only caught up at the very end, which suggests that in a real team setting, this would be a signal to either front-load simpler tasks earlier or improve mid-sprint check-ins to catch stalled progress before it compounds near the deadline.